

PROBLEM 01 · FINANCIAL CRIME

FRAUDMESH

Cross-Bank Financial Crime Reconstruction & Recovery Agent

THE PROBLEM

Modern financial fraud is increasingly a network problem rather than a single suspicious transaction. One scam can involve many victims, accounts and devices before the money leaves the system.

- multiple victims
- bank accounts and UPI IDs
- phone numbers and devices
- mule accounts and transfers
- withdrawals
- communications
- previous complaints

THE SYSTEMIC GAP

Existing systems can flag suspicious transactions or identifiers. The open question is different:

How do investigators reconstruct the complete fraud chain across fragmented incidents, and connect apparently unrelated cases?

DOCUMENTED FACT

- Between 2021 and 2025, India's National Cyber Crime Reporting Portal received more than 65.89 lakh financial-fraud complaints, with reported fraud exceeding INR 55,050 crore [5].
- In 2024 alone, 36,37,288 incidents and losses of INR 22,845.73 crore were reported, as stated by the Ministry of Home Affairs in Parliament [6].
- I4C's CFCFRMS reports more than INR 11,158 crore saved across 32.80 lakh complaints by 30 June 2026 [5].
- I4C launched a Suspect Registry of cybercriminal identifiers with banks on 10 September 2024 [6].

Figures are complaint-based government statements. They show scale and active national infrastructure for reporting; they do not by themselves measure how much investigators struggle to link cases.

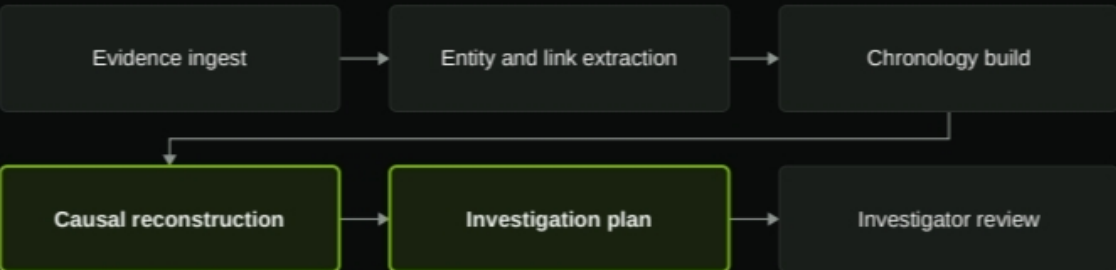
OUR PROPOSED SOLUTION

Build an investigation agent that ingests synthetic or authorized records, then reconstructs the chronology and relationship graph of a fraud operation.

- transaction histories
- complaint records
- communication metadata
- account relationships
- device identifiers
- timestamps
- previous incidents

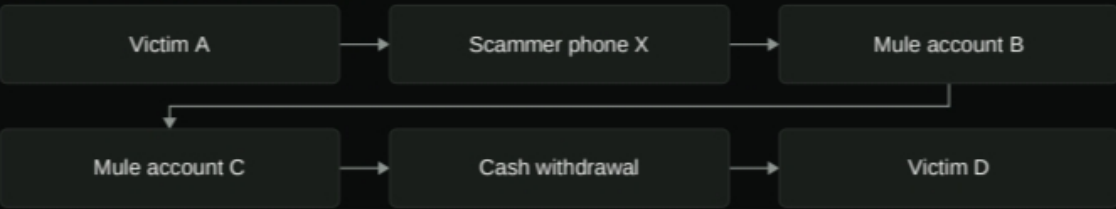
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SOLUTION PIPELINE



Green outline marks where Nemotron does the core reasoning.

EXAMPLE RECONSTRUCTED CHAIN



- common entities across complaints
- money-flow relationships
- temporal relationships
- contradictions between accounts
- missing evidence
- potentially related incidents
- next investigation steps

WHY NEMOTRON

The task is to move from evidence to relationships to chronology to causal reconstruction to an investigation plan. That is reasoning over a large body of interconnected evidence, not classifying one transaction.

Remove-Nemotron test. Exact identifier matches can be joined by a graph query, and that part needs no model. Nemotron is needed for free-text complaints, inconsistent narratives, inferred links without a shared identifier, and for writing a defensible plan. If the demo only joins on shared IDs, the idea fails this test.

HYPOTHETICAL DEMO SCENARIO

A synthetic dataset with 20 victims, 50+ transactions, 8 bank accounts, 6 phone numbers and multiple complaints. The cases initially look unrelated. Nemotron shows that several complaints belong to one fraud network.

ASSUMPTION limits and open points

- Synthetic or authorized data only. Real bank or I4C records are not assumed to be available.
- Outputs are investigative leads for human review, not accusations against any person.
- Entity resolution on messy real data is hard; the demo should show how uncertainty is flagged.